

Mode 11 — Baseline-BFP

C-DP16 (C8×C8) → DP32 (8×8) → DP16 (8×8) → DP16 (8×4) → DP8 (8×4) → block exponents

Step 1 — C-DP16 (C8×C8)

$$\text{C-DP16 (C8} \times \text{C8)} = D = \sum_{i=0}^{15} a_i b_i, \quad a_i = a_i^{re} + j a_i^{im}, \quad b_i = b_i^{re} + j b_i^{im}$$

Step 2 — DP32 (8×8)

$$\text{Re}(D) = \underbrace{\sum_{i=0}^{15} a_i^{re} b_i^{re} - \sum_{i=0}^{15} a_i^{im} b_i^{im}}_{\text{DP32 (8} \times \text{8)}}$$

$$\text{Im}(D) = \underbrace{\sum_{i=0}^{15} a_i^{re} b_i^{im} + \sum_{i=0}^{15} a_i^{im} b_i^{re}}_{\text{DP32 (8} \times \text{8)}}$$

Step 3 — DP16 (8×8)

$$\text{Re}(D) = \underbrace{\sum_{i=0}^{15} a_i^{re} b_i^{re}}_{\text{DP16 (8} \times \text{8)}} - \underbrace{\sum_{i=0}^{15} a_i^{im} b_i^{im}}_{\text{DP16 (8} \times \text{8)}}$$

$$\text{Im}(D) = \underbrace{\sum_{i=0}^{15} a_i^{re} b_i^{im}}_{\text{DP16 (8} \times \text{8)}} + \underbrace{\sum_{i=0}^{15} a_i^{im} b_i^{re}}_{\text{DP16 (8} \times \text{8)}}$$

Step 4 — DP16 (8×4)

$$\text{Re}(D) = 2^4 \underbrace{\sum_{i=0}^{15} a_i^{re} b_i^{re,hi}}_{\text{DP16 (8} \times \text{4)}} + \underbrace{\sum_{i=0}^{15} a_i^{re} b_i^{re,lo}}_{\text{DP16 (8} \times \text{4)}} - 2^4 \underbrace{\sum_{i=0}^{15} a_i^{im} b_i^{im,hi}}_{\text{DP16 (8} \times \text{4)}} - \underbrace{\sum_{i=0}^{15} a_i^{im} b_i^{im,lo}}_{\text{DP16 (8} \times \text{4)}}$$

$$\text{Im}(D) = 2^4 \underbrace{\sum_{i=0}^{15} a_i^{re} b_i^{im,hi}}_{\text{DP16 (8} \times \text{4)}} + \underbrace{\sum_{i=0}^{15} a_i^{re} b_i^{im,lo}}_{\text{DP16 (8} \times \text{4)}} + 2^4 \underbrace{\sum_{i=0}^{15} a_i^{im} b_i^{re,hi}}_{\text{DP16 (8} \times \text{4)}} + \underbrace{\sum_{i=0}^{15} a_i^{im} b_i^{re,lo}}_{\text{DP16 (8} \times \text{4)}}$$

Step 5 — DP8 (8×4)

$$\begin{aligned} \text{Re}(D) = & 2^4 \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{re,hi}}_{\text{DP8 (8} \times \text{4)}} + 2^4 \underbrace{\sum_{i=8}^{15} a_i^{re} b_i^{re,hi}}_{\text{DP8 (8} \times \text{4)}} + \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{re,lo}}_{\text{DP8 (8} \times \text{4)}} + \underbrace{\sum_{i=8}^{15} a_i^{re} b_i^{re,lo}}_{\text{DP8 (8} \times \text{4)}} \\ & - 2^4 \underbrace{\sum_{i=0}^7 a_i^{im} b_i^{im,hi}}_{\text{DP8 (8} \times \text{4)}} - 2^4 \underbrace{\sum_{i=8}^{15} a_i^{im} b_i^{im,hi}}_{\text{DP8 (8} \times \text{4)}} - \underbrace{\sum_{i=0}^7 a_i^{im} b_i^{im,lo}}_{\text{DP8 (8} \times \text{4)}} - \underbrace{\sum_{i=8}^{15} a_i^{im} b_i^{im,lo}}_{\text{DP8 (8} \times \text{4)}} \end{aligned}$$

$$\begin{aligned} \text{Im}(D) = & 2^4 \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{im,hi}}_{\text{DP8 (8} \times \text{4)}} + 2^4 \underbrace{\sum_{i=8}^{15} a_i^{re} b_i^{im,hi}}_{\text{DP8 (8} \times \text{4)}} + \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{im,lo}}_{\text{DP8 (8} \times \text{4)}} + \underbrace{\sum_{i=8}^{15} a_i^{re} b_i^{im,lo}}_{\text{DP8 (8} \times \text{4)}} \\ & + 2^4 \underbrace{\sum_{i=0}^7 a_i^{im} b_i^{re,hi}}_{\text{DP8 (8} \times \text{4)}} + 2^4 \underbrace{\sum_{i=8}^{15} a_i^{im} b_i^{re,hi}}_{\text{DP8 (8} \times \text{4)}} + \underbrace{\sum_{i=0}^7 a_i^{im} b_i^{re,lo}}_{\text{DP8 (8} \times \text{4)}} + \underbrace{\sum_{i=8}^{15} a_i^{im} b_i^{re,lo}}_{\text{DP8 (8} \times \text{4)}} \end{aligned}$$

Step 6 — Block exponents (one exponent block per DP8 (8×4); Re/Im share E_k)

$$M^{re} = 2^{-\delta_0} \left(2^4 \sum_{i=0}^7 a_i^{re} b_i^{re,hi} + \sum_{i=0}^7 a_i^{re} b_i^{re,lo} - 2^4 \sum_{i=0}^7 a_i^{im} b_i^{im,hi} - \sum_{i=0}^7 a_i^{im} b_i^{im,lo} \right) + 2^{-\delta_1} \left(\dots i \in [8, 15] \dots \right)$$

$$M^{im} = 2^{-\delta_0} \left(2^4 \sum_{i=0}^7 a_i^{re} b_i^{im,hi} + \sum_{i=0}^7 a_i^{re} b_i^{im,lo} + 2^4 \sum_{i=0}^7 a_i^{im} b_i^{re,hi} + \sum_{i=0}^7 a_i^{im} b_i^{re,lo} \right) + 2^{-\delta_1} \left(\dots i \in [8, 15] \dots \right)$$

$E_k = E_{a,k} + E_{b,k}$; $E = \max(E_0, E_1)$, $\delta_k = E - E_k$; truncating shifts, on block sums only; $\text{Re}(\hat{D}) = M^{re} \cdot 2^E$, $\text{Im}(\hat{D}) = M^{im} \cdot 2^E$.

Step 7 — Result

$$M^{re} = 2^{-\delta_0} \left(2^4 \sum_{i=0}^7 a_i^{re} b_i^{re,hi} + \sum_{i=0}^7 a_i^{re} b_i^{re,lo} - 2^4 \sum_{i=0}^7 a_i^{im} b_i^{im,hi} - \sum_{i=0}^7 a_i^{im} b_i^{im,lo} \right)$$

$$+ 2^{-\delta_1} \left(2^4 \sum_{i=8}^{15} a_i^{re} b_i^{re,hi} + \sum_{i=8}^{15} a_i^{re} b_i^{re,lo} - 2^4 \sum_{i=8}^{15} a_i^{im} b_i^{im,hi} - \sum_{i=8}^{15} a_i^{im} b_i^{im,lo} \right)$$

$$M^{im} = 2^{-\delta_0} \left(2^4 \sum_{i=0}^7 a_i^{re} b_i^{im,hi} + \sum_{i=0}^7 a_i^{re} b_i^{im,lo} + 2^4 \sum_{i=0}^7 a_i^{im} b_i^{re,hi} + \sum_{i=0}^7 a_i^{im} b_i^{re,lo} \right)$$

$$+ 2^{-\delta_1} \left(2^4 \sum_{i=8}^{15} a_i^{re} b_i^{im,hi} + \sum_{i=8}^{15} a_i^{re} b_i^{im,lo} + 2^4 \sum_{i=8}^{15} a_i^{im} b_i^{re,hi} + \sum_{i=8}^{15} a_i^{im} b_i^{re,lo} \right),$$

$$E = \max(E_0, E_1)$$